PocketSmart AI: Your Smart Budget & Recommendation Assistant

Project Description

PocketSmart AI uses Generative AI to turn a budget into a personalised, shop-able plan for **home interiors, parties and jewelry**. A FastAPI backend sends validated user input (text, numbers and an optional outfit image) to Google Gemini, then re-checks the answer so the plan never exceeds the budget, and attaches search links for Amazon, Flipkart, IKEA, Swiggy, Zomato, OYO and other platforms. Users register, sign in, and revisit saved plans from their history.

Scenarios

Scenario 1: Home Interior Planning. The user opens the Home Budget Planner, enters a budget, quantities of lights, ceiling fans, furniture and dining tables, and selects rooms. Gemini recommends cost-effective options for each category across IKEA, Amazon, Flipkart, Myntra and Ajio, balanced for function, style and price.

Scenario 2: AI-Based Party Budget Planning. In the Party Planner the user enters budget, guest count, party type and venue. The budget is allocated across venue, catering, decoration and entertainment, with links to Swiggy, Zomato, BookMyShow, OYO and more, plus a contingency amount.

Scenario 3: Jewelry Recommendations. The Jewelry Planner takes a budget, occasion and style preferences, and optionally an outfit photo. Gemini analyses colours and formality and suggests matching pieces from Amazon, Flipkart, BlueStone, Tanishq, CaratLane and others.

Technical Architecture

Three tiers: a responsive HTML/CSS/JS front end (Jinja2 templates), a FastAPI back end (routes, JWT sessions, validation, budget engine) and external services (Gemini API, shop search URLs) with SQLite for storage. See *3. Project Design Phase / Solution Architecture* for the diagram.

Pre-requisites

1. Python 3.10+ - https://www.python.org
2. FastAPI - https://fastapi.tiangolo.com
3. Google AI Studio account and Gemini API key - https://aistudio.google.com/apikey
4. Git and a GitHub account
5. Basic HTML, CSS and JavaScript

Project Workflow

Milestone	Activities	Where in the code
1. Gemini AI Initialization	1.1 Create Google AI Studio account; 1.2 generate and store the API key in `.env`; 1.3 validate text and image connectivity	`.env.example`, `config.py`, `tools/check_gemini.py`
2. Core Functionalities Development	2.1 FastAPI initialization and imports; 2.2 planner endpoints with Gemini prompts; 2.3 login/register routing; 2.4 modular structure with `/token`, `/session-info`, `/session-data`	`main.py`, `gemini_utils.py`, `auth.py`, `models.py`
3. Backend - FastAPI Integration	3.1 `/home-budget`, `/party-budget`, `/jewelry-budget` (aliases `/generate-*`) and auth routes; 3.2 `/recommendation-details/{id}`; 3.3 CORS, `/history`; 3.4 startup task and `__main__`	`main.py`
4. UI Development	4.1 landing page, forms, card-style results; dynamic rendering with JavaScript	`templates/`, `static/`
5. Testing & Optimization	5.1 real-world inputs across the planners; 5.2 response quality and budget adherence; 5.3 prompt tuning and validation; 5.4 fallback recommendations	`tests/`, `tools/perf_load.py`, `gemini_utils.py`

Milestone 1: Gemini AI Initialization

1. Sign in at https://aistudio.google.com and choose **Get API key** then **Create API key**.
2. Copy the key into `.env` as `GOOGLE_API_KEY` (never commit `.env`).
3. Run `python tools/check_gemini.py`. It sends a text prompt and a text + image prompt, tries the configured model and then fallback models, and prints which model to use.

Note: the original brief refers to "Gemini 1.5 Flash Pro". That model name does not exist and the 1.5 family is retired, so the model is configurable (`GEMINI_MODEL`).

Milestone 2 and 3: Backend

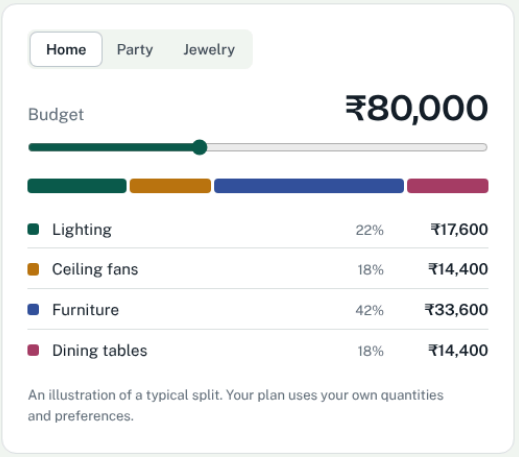
- `config.py` loads settings; `models.py` validates inputs; `database.py` stores users and plans; `auth.py` handles bcrypt, JWT and sessions.
- `gemini_utils.py` builds one prompt per planner, calls Gemini (JSON mode), sanitises the reply, recomputes totals, trims over-budget items, builds shopping links and falls back to default plans if needed.
- `main.py` exposes the routes: `/register`, `/login`, `/token`, `/logout`, `/session-info`, `/session-data`, `/home-planner`, `/party-planner`, `/jewelry-planner`, `/home-budget`, `/party-budget`, `/jewelry-budget`, `/recommendation-history`, `/recommendation-details/{id}`, `/history`, `/dashboard`, `/health`. Interactive API docs are at `/docs`.

Milestone 4: UI Development

Know where every rupee goes before you spend it.

Enter a budget for your home, a party or a piece of jewelry. PocketSmart splits it across what you need, checks the total, and links each item to the stores that sell it.

Create a free account | Sign in



An illustration of a typical split. Your plan uses your own quantities and preferences.

Three planners, one account

Each planner asks only for what it needs.

Home interior	Lights, ceiling fans, furniture and dining tables for the rooms you choose, with options from IKEA, Amazon, Flipkart, Myntra and Ajio.	Budget, quantities, rooms
Party	Venue, catering, decoration and entertainment for your guest count, with a contingency buffer. Links to Swiggy, Zomato, BookMyShow, OYO and more.	Budget, guests, event, venue
Jewelry	Pieces matched to the occasion and, if you add a photo, to the colours and style of your outfit. Links to BlueStone, Tanishq, CaratLane, Amazon and more.	Budget, occasion, optional photo

How it works

1

Enter your budget

Choose a planner and fill in a short form. Quick amounts save typing.

2

Get a checked plan

Gemini proposes the items. The server rechecks every price and removes anything that would take you over budget.

3

Shop and come back

Open the store links for each item. Every plan is saved in your history.

The AI suggests. The server verifies.

Totals you can trust

Planned spend and remaining budget are calculated by the server, never taken from the AI's own sums.

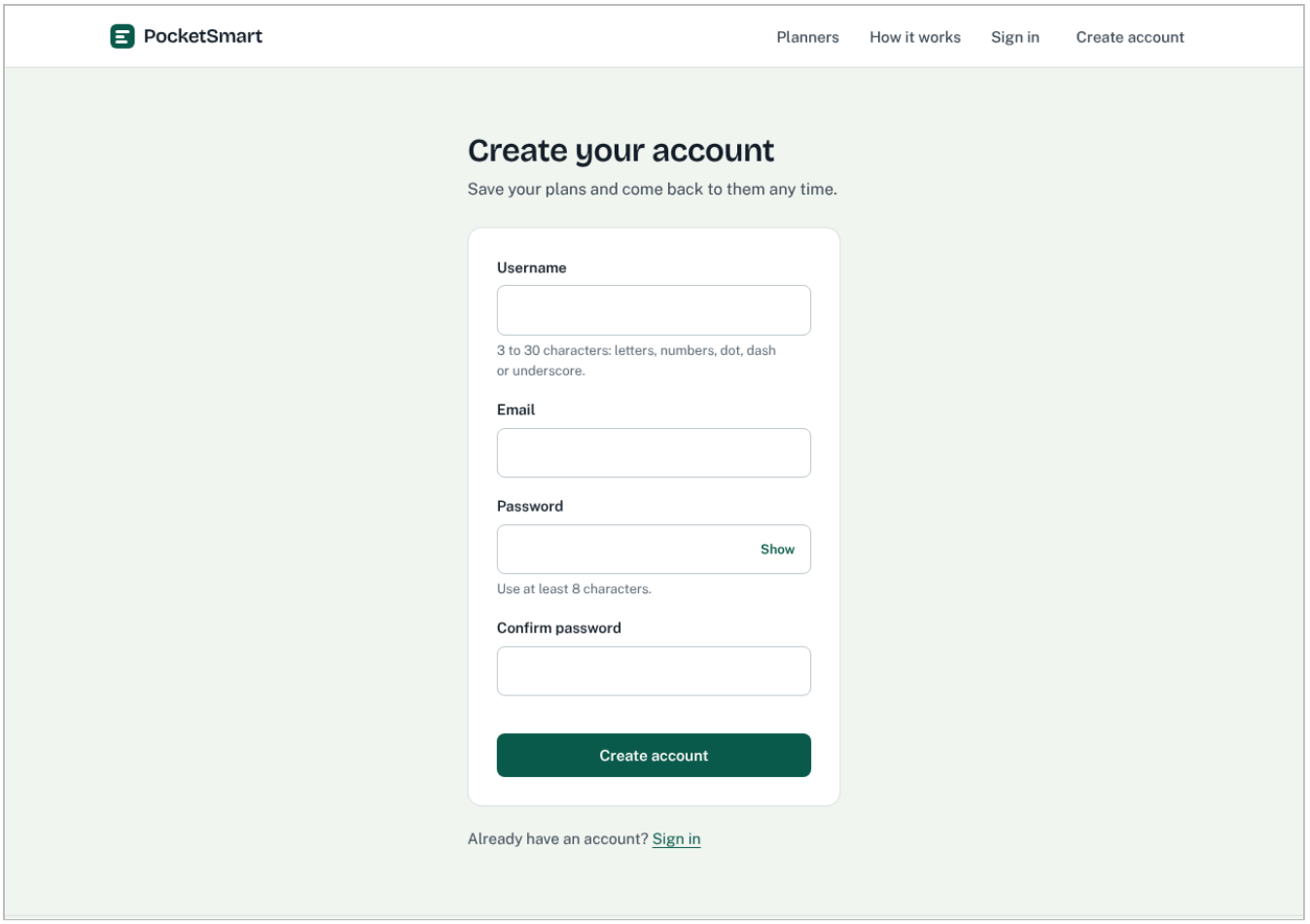
Always an answer

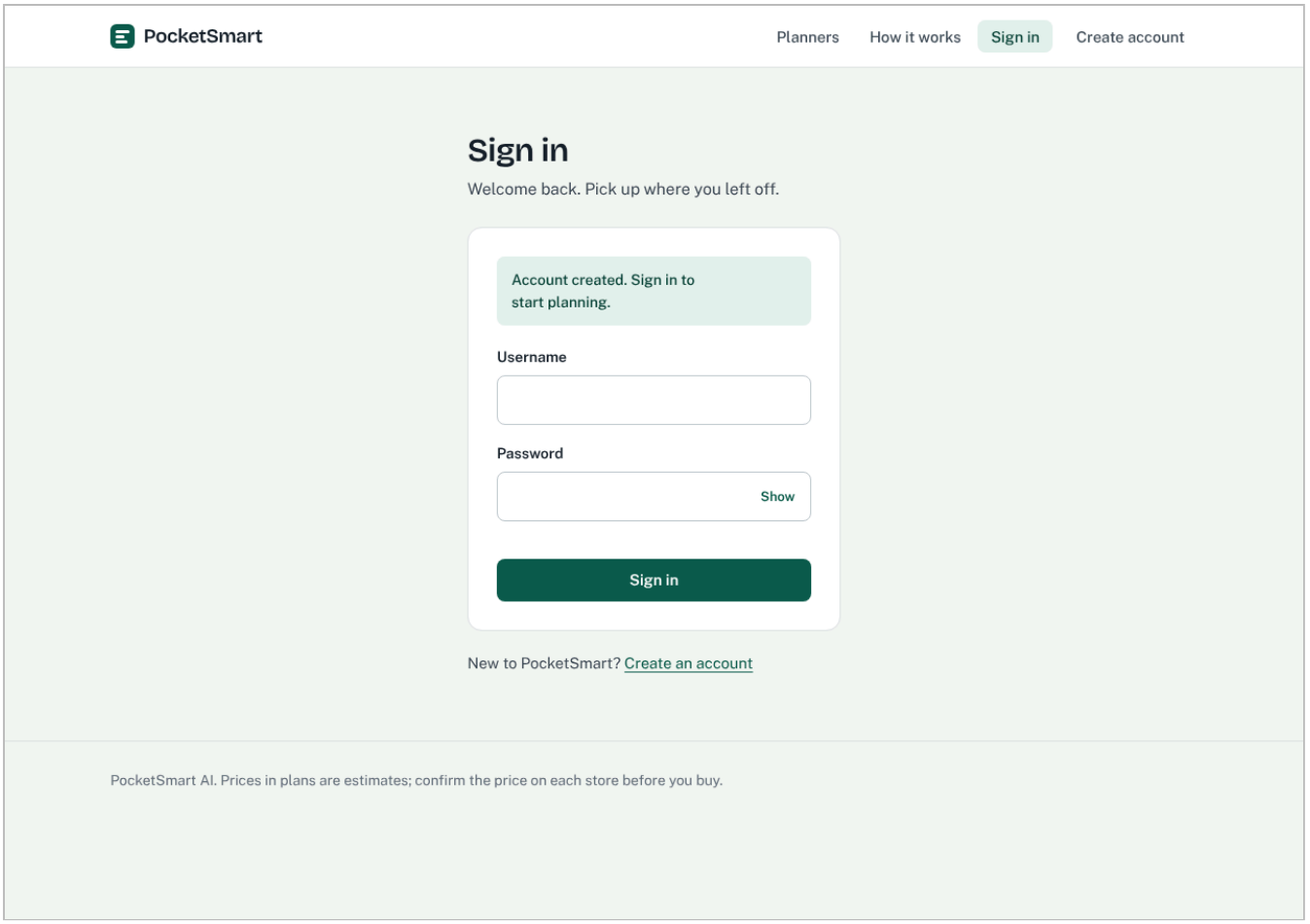
If the AI service is unavailable, you get a clearly labelled standard estimate instead of an error.

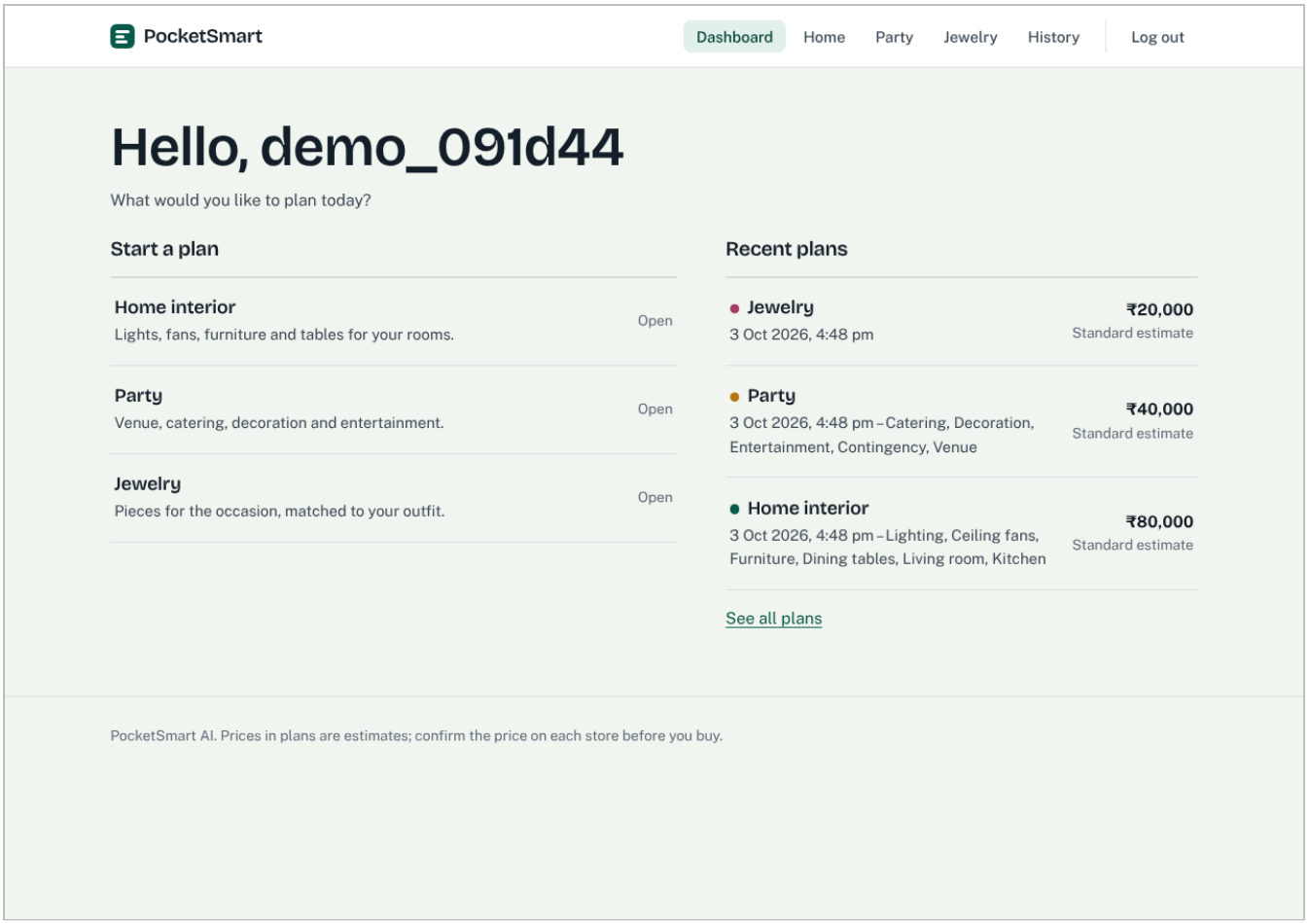
Your plans stay yours

Passwords are hashed, sessions expire after 30 minutes, and only you can open your saved plans.

PocketSmart AI. Prices in plans are estimates; confirm the price on each store before you buy.




PocketSmart
Planners
How it works
Sign in
Create account
Sign in
Welcome back. Pick up where you left off.
Account created. Sign in to start planning.
Username
Password
Show
Sign in
New to PocketSmart? Create an account
PocketSmart AI. Prices in plans are estimates; confirm the price on each store before you buy.


PocketSmart
Dashboard
Home
Party
Jewelry
History
Log out
Hello, demo_091d44
What would you like to plan today?
Start a plan
Home interior
Lights, fans, furniture and tables for your rooms.
Open
Party
Venue, catering, decoration and entertainment.
Open
Jewelry
Pieces for the occasion, matched to your outfit.
Open
Recent plans
Jewelry
3 Oct 2026, 4:48 pm
₹20,000
Standard estimate
Party
3 Oct 2026, 4:48 pm – Catering, Decoration, Entertainment, Contingency, Venue
₹40,000
Standard estimate
Home interior
3 Oct 2026, 4:48 pm – Lighting, Ceiling fans, Furniture, Dining tables, Living room, Kitchen
₹80,000
Standard estimate
See all plans
PocketSmart AI. Prices in plans are estimates; confirm the price on each store before you buy.

Milestone 5: Testing & Optimization

Home Interior Planner and recommendations

Home interior planner

Tell us your budget and what the home needs. We split the money across lighting, fans, furniture and tables.

Budget

Total budget

₹ 80,000

The most you want to spend in total.

₹25,000 ₹50,000 ₹1,00,000

What you need

Lights	Ceiling fans
− 5 +	− 2 +

Furniture pieces	Dining tables
− 2 +	− 1 +

Rooms

Living room | Kitchen | Bedroom

Preferences

Anything else we should know

Minimalist, warm lighting

Build my plan

Home interior plan

Print plan

Saved to your history just now.

The AI service was unavailable, so this is a standard estimate based on typical prices. Try again later for a tailored plan.

Budget	Planned	Left over
₹80,000	₹42,950	₹37,050

Lighting	3%	₹2,250
Ceiling fans	10%	₹7,600
Furniture	6%	₹4,800
Dining tables	16%	₹12,500
Living room	18%	₹14,000
Kitchen	2%	₹1,800
Unspent	46%	₹37,050

Lighting — ₹2,250 of ₹2,250

LED batten / panel light — ₹2,250 (5 × ₹450)

Energy-efficient 20W LED light, cool or warm white.

Search on: Amazon | Flipkart | IKEA | Myntra | Ajio

Ceiling fans — ₹7,600 of ₹7,600

BLDC ceiling fan 1200mm — ₹7,600 (2 × ₹3,800)

Energy-saving BLDC ceiling fan with remote.

Search on: Amazon | Flipkart | IKEA | Myntra | Ajio

Furniture — ₹4,800 of ₹4,800

Engineered-wood chair — ₹4,800 (2 × ₹2,400)

Sturdy, easy-to-assemble chair for living areas.

Search on: Amazon | Flipkart | IKEA | Myntra | Ajio

Dining tables — ₹12,500 of ₹12,500

4-seater dining table set — ₹12,500

Compact 4-seater dining table with chairs.

Search on: Amazon | Flipkart | IKEA | Myntra | Ajio

Living room — ₹14,000 of ₹14,000

3-seater fabric sofa — ₹14,000

Comfortable sofa for the living room.

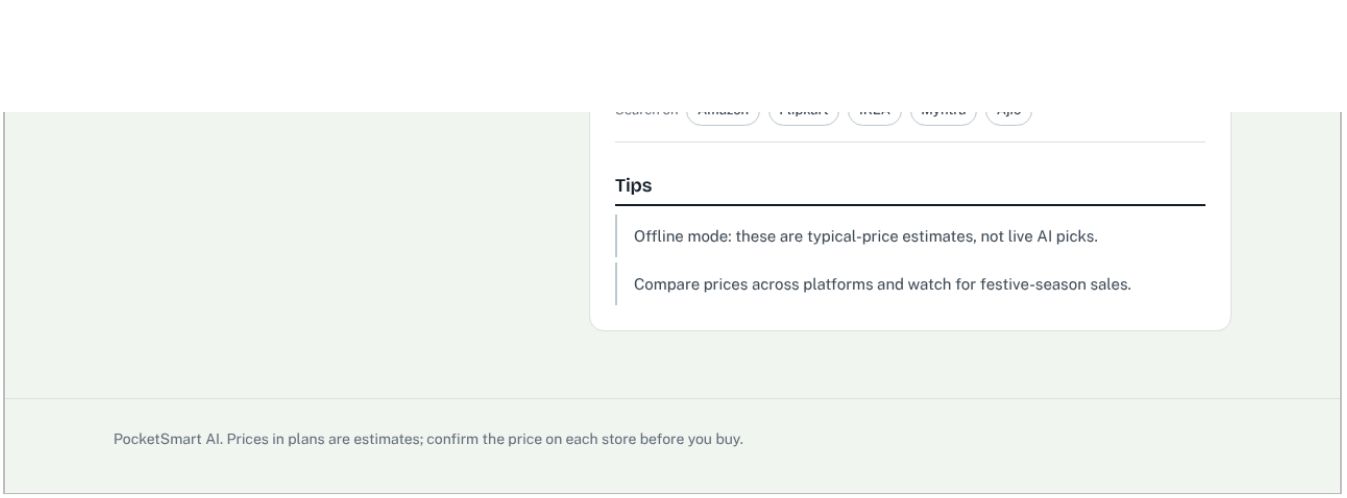
Search on: Amazon | Flipkart | IKEA | Myntra | Ajio

Kitchen — ₹1,800 of ₹1,800

Modular kitchen storage rack — ₹1,800

Stainless-steel rack for utensils and spices.

Search on: Amazon | Flipkart | IKEA | Myntra | Ajio



Party Planner and recommendations

Party planner

Give us the budget, guest count and event. We divide it between venue, catering, decoration and entertainment, and keep a small buffer.

Budget and guests

Total budget

₹ 40,000

₹20,000 ₹50,000 ₹1,00,000

Number of guests

30

Event

Party type: Birthday

Venue: Banquet hall

What to plan for

Catering Decoration Entertainment

Preferences

Anything else we should know

Vegetarian menu, a jungle theme, music for all ages

Build my plan

Party plan

Print plan

Saved to your history just now.

The AI service was unavailable, so this is a standard estimate based on typical prices. Try again later for a tailored plan.

Budget	Planned	Left over
₹40,000	₹37,997	₹2,003

Catering	36%	₹14,490
Decoration	14%	₹5,796
Entertainment	14%	₹5,474
Contingency	6%	₹2,576
Venue	24%	₹9,661
Unspent	5%	₹2,003

Catering — ₹14,490 of ₹14,490

Birthday - Catering per guest — ₹14,490

Veg meal / snacks per guest via Swiggy or Zomato party packs.

30 × ₹483

Search on: Swiggy, Zomato

Decoration — ₹5,796 of ₹5,796

Birthday - Decoration kit — ₹5,796

Balloons, banners, lights and table decor for the theme.

Search on: Amazon, Flipkart, Meesho, Myntra

Entertainment — ₹5,474 of ₹5,474

Birthday - Entertainment and games — ₹5,474

Music playlist, speaker rental and group games.

Search on: BookMyShow, Amazon, Flipkart

Contingency — ₹2,576 of ₹2,576

Birthday - Contingency buffer — ₹2,576

Reserve for last-minute needs.

Search on: Amazon, Flipkart, Google

Venue — ₹9,661 of ₹9,661

Birthday - Venue hire — ₹9,661

Banquet hall or terrace space for your guest count.

Search on: Google, Booking, MakeMyTrip, OYO, NoBroker

Venue ideas

Birthday party hall — ₹9,661

Banquet hall. up to 30 guests

Search on: Google, Booking, MakeMyTrip, OYO, NoBroker

Tips

Offline mode: allocation follows a standard party-budget split.

Confirm vendor quotes before booking.

PocketSmart AI. Prices in plans are estimates; confirm the price on each store before you buy.

Jewelry Planner (with outfit image) and recommendations

PocketSmart

Dashboard | Home | Party | Jewelry | History | Log out

Jewelry planner

Pick an occasion and a budget. Add a photo of your outfit and the AI matches jewelry to its colours and style.

Budget

Total budget

₹ 20,000

₹10,000 ₹25,000 ₹50,000

Occasion and style

Occasion

Wedding

Style preferences

Gold, traditional

Outfit photo (optional)

_outfit.png (2 KB). Choose a different photo to replace it.



Remove photo

Build my plan

Jewelry plan

Print plan

Saved to your history just now.

The AI service was unavailable, so this is a standard estimate based on typical prices. Try again later for a tailored plan.

Budget	Planned	Left over
₹20,000	₹19,000	₹1,000

Recommended pieces

Necklace ₹9,000

A traditional gold-plated necklace suited to a wedding. traditional gold-plated

Search on Amazon Flipkart BlueStone Tanishq CaratLane Melorra Meesho

Earrings ₹6,000

A traditional gold-plated earrings suited to a wedding. traditional gold-plated

Search on Amazon Flipkart BlueStone Tanishq CaratLane Melorra Meesho

Bracelet ₹4,000

A traditional gold-plated bracelet suited to a wedding. traditional gold-plated

Search on Amazon Flipkart BlueStone Tanishq CaratLane Melorra Meesho

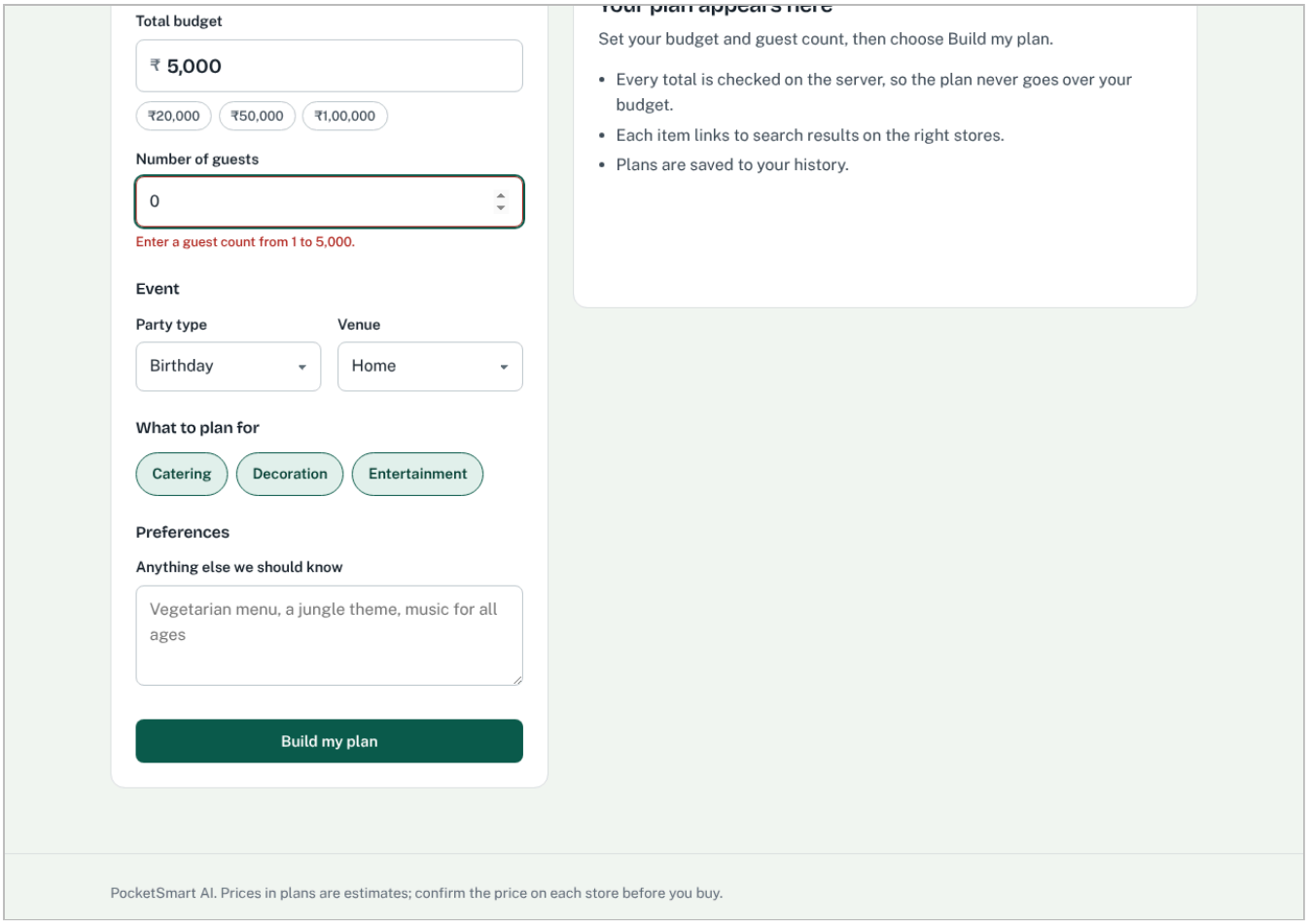
Tips

Offline mode: outfit photo analysis needs the Gemini service.

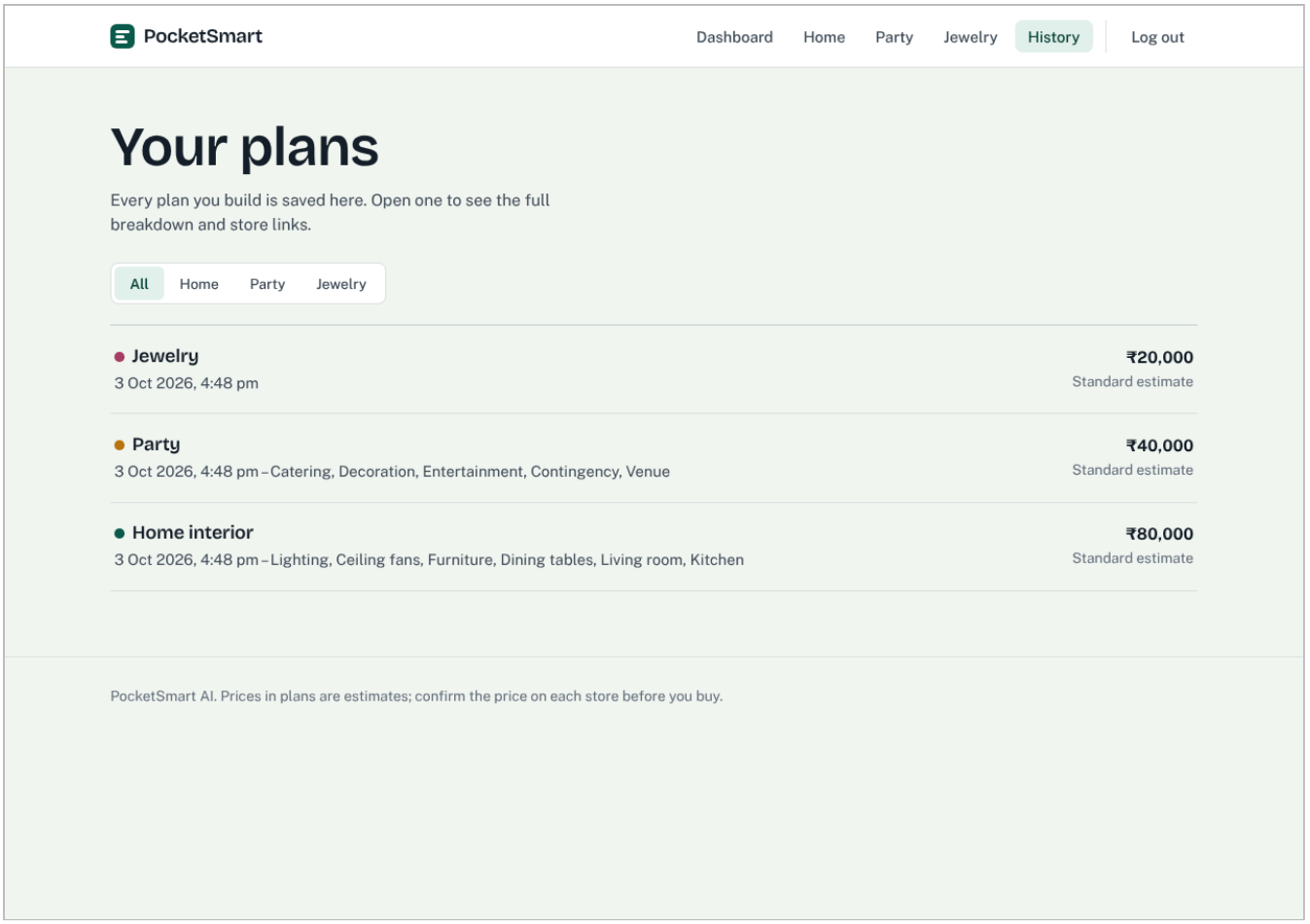
Match metal tone (gold/silver) across all pieces.

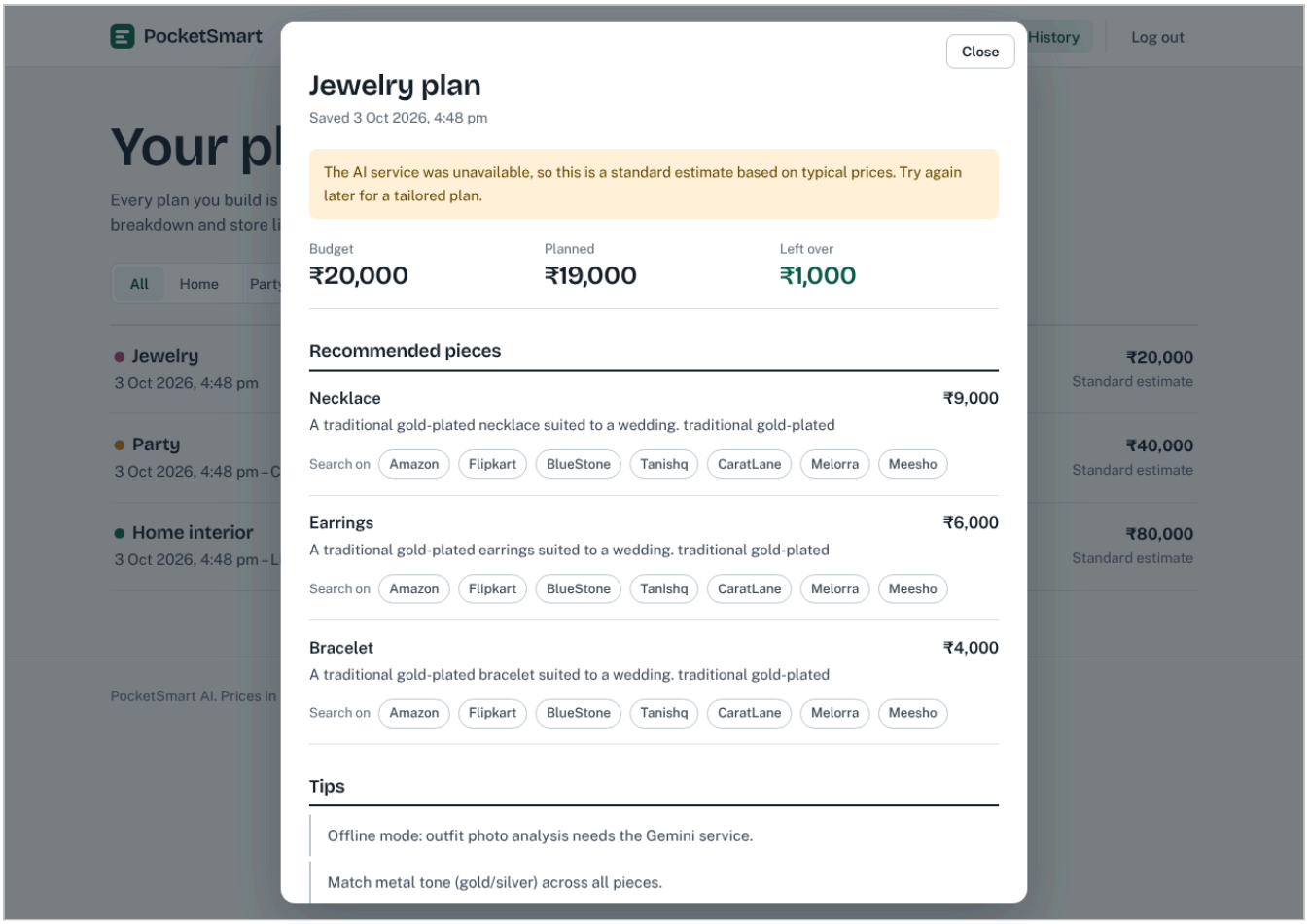
PocketSmart AI. Prices in plans are estimates; confirm the price on each store before you buy.

Input validation



Recommendation history and details


PocketSmart
Dashboard
Home
Party
Jewelry
History
Log out
Your plans
Every plan you build is saved here. Open one to see the full breakdown and store links.
All
Home
Party
Jewelry
Jewelry
3 Oct 2026, 4:48 pm
₹20,000
Standard estimate
Party
3 Oct 2026, 4:48 pm – Catering, Decoration, Entertainment, Contingency, Venue
₹40,000
Standard estimate
Home interior
3 Oct 2026, 4:48 pm – Lighting, Ceiling fans, Furniture, Dining tables, Living room, Kitchen
₹80,000
Standard estimate
PocketSmart AI. Prices in plans are estimates; confirm the price on each store before you buy.


PocketSmart
History
Log out
Close
Jewelry plan
Saved 3 Oct 2026, 4:48 pm
The AI service was unavailable, so this is a standard estimate based on typical prices. Try again later for a tailored plan.
Budget
₹20,000
Planned
₹19,000
Left over
₹1,000
Recommended pieces
Necklace
₹9,000
A traditional gold-plated necklace suited to a wedding. traditional gold-plated
Search on
Amazon
Flipkart
BlueStone
Tanishq
CaratLane
Melorra
Meesho
Earrings
₹6,000
A traditional gold-plated earrings suited to a wedding. traditional gold-plated
Search on
Amazon
Flipkart
BlueStone
Tanishq
CaratLane
Melorra
Meesho
Bracelet
₹4,000
A traditional gold-plated bracelet suited to a wedding. traditional gold-plated
Search on
Amazon
Flipkart
BlueStone
Tanishq
CaratLane
Melorra
Meesho
Tips
Offline mode: outfit photo analysis needs the Gemini service.
Match metal tone (gold/silver) across all pieces.

These screenshots were captured by the automated browser test `tools/capture_screenshots.py` running **in offline-fallback mode** (no API key), which is why results are labelled "Offline estimate". With a Gemini key the label reads "Gemini AI" and the items come from the model.

Conclusion

PocketSmart AI combines FastAPI with Google Gemini to give budget-aware recommendations for home interiors, parties and jewelry. The server verifies every plan against the budget, falls back gracefully if the AI is unavailable, protects user data with hashed passwords and signed sessions, and links each item to the platforms where it can be bought.